

TOUCHSCREEN PANELS COMPARISON

Technology	Resistive live-wire	Capacitive surface	Capacitive projected	Infrared	SAW
Performance:					
Response time	<15ms	<15ms	<15ms	<20ms	<10 ms
Transmission/clarity	75% - 85%	90% - 98%	90% - 98%	95% - 100%	90% - 98%
Multi-touch	Limited	No	Yes	Yes	No
High sensitivity (light touch)	Good	Very good	Very good	Best	Very good
Input flexibility:					
Finger	Yes	Yes	Yes	Yes	Yes
Glove	Yes	No	Yes	Yes	Yes
Stylus	Yes	Capacitive stylus	Capacitive stylus surgical glove	Yes	Yes
Soft-tip stylus	Yes	No	No	Yes	Yes
Electrical:					
Immunity to surrounding electric or magnetic fields	Best	Best	Best	Best	Very good
Low emissions	Best	Best	Best	Best	Best
Environmental:					
Non-sensitivity to humidity	Best	Best	Best	Very good	Very good
Shock, vibration	Best	Very good	Very good	Best	Very good
Chemical resistance	Very good	Very good	Best	Best	Very good
Durability, wear	Low	Medium	High	High	Low
Scratch resistance	Poor	Very good	Best	Best	Best
Resistive to surface contaminants	Best	Good	Very good	Good	Average

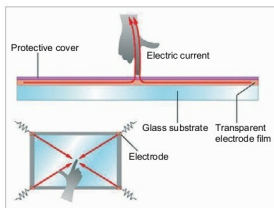


Fig. 3: Contact reaction to a surface-capacitive touch panel

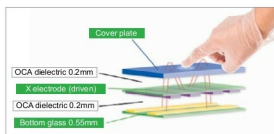


Fig. 4: The different layers

gloves, it is likely that the controller will not be able to register a touch point.

IR grid technology. Another type of mostly used touch panels are the IR touch panels (ITPs). IR technology consists of IR LEDs and light sensors that are placed on the perimeter of the bezel. LEDs transmit beams of light that run side by side to the assigned light sensor and create a light grid. If anything disrupts the light or reduces it, a touch point is registered by the controller.

Performance is sensitive to water, snow, rain and ambient light interference. The screen could malfunction due to build

up of dust, oil or grease. Cost is relatively high as compared to other touch technologies.

SAW touchscreen. A SAW is a structure-borne sound wave that spreads across a surface in a planar manner, that is, in two dimensions. The receiving transducer locates the touch point and sends this data to the controller.

SAW touchscreens have excellent image clarity, so can display high-detail graphics. These can be activated by a finger, gloved hand or soft-tip stylus.

However, SAW panels are the most expensive of the three liquids, and any contact with something hard like a pen or finger nail can cause a false trigger. Solid contaminants on the screen can create non-touch areas, until removed.

Optical touchscreens. This is a relatively-modern development in touchscreen technology, which uses two line-scanning cameras located at the corners of the screen. IR backlights are placed in the camera's field-of-view on the other side of the screen.

This technology is growing in popularity due to its scalability, versatility and affordability, especially for bigger units. It permits 100 per cent light transmission and requires only one calibration.

Near-field imaging touchscreens. This is one of the newest technologies and consists of two touchscreens. When the finger, glove or any other conductive stylus comes into contact with the sensor, electrostatic field is disturbed. It is an extremely-durable screen that is suited for use in industrial control systems and other harsh environments.

Factors to consider for buying touchscreen panels

Selection of a touchscreen panel is basically determined by the application and expected users. A panel for a classroom (teacher and students) will have different quali-